LAB WORKSHEET

Security Labs

A set of self-contained cyber ranges: reconnaissance, auditing, incident response, mobile forensics and SQL injection. Everything is fully simulated inside the page.

Ethics & safety first. Every target, host, hash and database in Security Labs is **fictional and simulated inside your browser**. Nothing here touches a real system or network. The skills are taught for **defence and authorised testing only** — never use them against systems you do not own or have explicit written permission to test.

The ranges

1. **Range** — reconnaissance and enumeration against a simulated host: scan, identify services, map what's exposed.
2. **Audit** — review a system's configuration and spot the weaknesses (weak passwords, open services, missing patches).
3. **Incident** — work a simulated incident: read the signs, contain, and record what happened.
4. **Mobile forensics** — examine an acquired phone image: recover artefacts and answer the case questions.
5. **SQL injection & prevention** — see how an injection bypasses a login against a fake in-page table, then learn the fix (parameterised queries, input validation).
6. **Free play & capstone** — open-ended practice and a combined challenge.

How to work a range

1. Read the brief at the top of each tab — it sets the goal and the rules of engagement.
2. Use the simulated tools provided; type `help` where a console is offered.
3. Answer the case questions to mark objectives complete.

SQL injection — the lesson

A classic login bypass works because input is concatenated straight into the query. Entering something like `' OR '1'='1` makes the WHERE clause always true.

```
-- vulnerable:
SELECT * FROM users WHERE name='$input';
-- safe (parameterised):
SELECT * FROM users WHERE name = ?;
```

Prevention: parameterised queries / prepared statements, input validation, least-privilege database accounts, and never trusting client input.

Record your findings on the page whiteboard as you go — it's a good habit for real incident notes.